Gradient flow

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1 Vanilla gradient and the improvement

Consider the result of some ML algorithms to be $x_t \in \mathbb{R}^n$, t is for time step, and choose $f: \mathbb{R}^n \rightarrow \mathbb{R}$ to be the differtiable loss function, since we want loss function to decrease by time, which is $\frac{d f(x_t)}{\mathrm{dt}} \leqslant 0 \Rightarrow 0 \leqslant \left\langle -\nabla f(x_t), \frac{d x_t}{\mathrm{dt}} \right\rangle \leqslant \|-\nabla f(x_t)\| \left\| \frac{d x_t}{\mathrm{dt}} \right\|$.The Cauchy-Schwarz takes equality iff:

$$\frac{d x_t}{\mathrm{dt}} = -\eta \nabla f(x_t). \tag{1}$$

Use $\eta = 1$, $x_{t+1} - x_t = -\gamma \nabla f(x_t)$, which is the vanilla gradient descent.

By the word of Jianlin Su, SGD and GD have no remarkable difference, so we back to think about the dynamic changing of (1).

We think about:

$$\underbrace{\frac{d^2 x_t}{\mathrm{dt}^2}}_{a} + \lambda \frac{d x_t}{\mathrm{dt}} = -\nabla f(x_t), \lambda > 0, \tag{2}$$

we find both (1) and (2) have the same fix point $x_t = 0$.

Intuitively, Newton's 2nd law is about the force and accelaration, which is also the 2nd derivative of displacement. So this equation can be explained by N2 with dumped force.

Use central-difference to difference can get $O(\gamma^2)$ precision:

$$\frac{x_{t+1} - 2x_t + x_{t-1}}{\gamma^2} + \lambda \frac{x_{t+1} - x_{t-1}}{2\gamma} = -\nabla f(x_t). \tag{3}$$

choose $v_{t+1} = x_{t+1} - x_t$, we get:

$$v_{t+1} = \frac{\frac{1}{\gamma^2} - \frac{\lambda}{2\gamma}}{\frac{1}{\gamma^2} + \frac{\lambda}{2\gamma}} v_t - \frac{\nabla f(x_t)}{\frac{1}{\gamma^2} + \frac{\lambda}{2\gamma}} \tag{4}$$

Choose $\beta := \frac{\frac{1}{\gamma^2} - \frac{\lambda}{2\gamma}}{\frac{1}{\gamma^2} + \frac{\lambda}{2\gamma}}, \alpha = \frac{1}{\gamma^2} + \frac{\lambda}{2\gamma}$:

$$\begin{cases} v_{t+1} = \beta v_n - \alpha \nabla f(x_t), \\ x_{t+1} = v_{t+1} + x_t. \end{cases} \tag{5}$$

This is also the GD with Momentum.

Remark 1. We can see MGD has step $\gamma \approx \sqrt{\alpha}$. Since $\gamma \ll 1 \Rightarrow \sqrt{\alpha} > \alpha$. So MGD has more precision but also large step.

Remark 2. Consider $\gamma \rightarrow 0, \sqrt{\alpha} \rightarrow \gamma, \beta \approx 1 - \frac{\lambda \sqrt{\alpha}}{2} \approx 1 - \lambda \sqrt{\alpha}$. Since quite small, 0.5 is not necessary.

Also, choose to find the closed form $\gamma(\alpha)$ and put back to β can get the same result.

Nesterov Momentum:

Apart from explicit form, we also have implicit form for discrete:

$$\frac{x_{t+1} - 2x_t + x_{t-1}}{\gamma^2} = -\lambda \frac{x_t - x_{t-1}}{\gamma} - \nabla f\left(\frac{x_{t+1} + x_t}{2}\right), \tag{6}$$

Still by $v_{t+1} = x_{t+1} - x_t$, $\beta = 1 - \lambda\gamma, \alpha = \gamma^2$:

$$\begin{cases} v_{t+1} = \beta v_t - \alpha \nabla f\left(\frac{x_{t+1} + x_t}{2}\right) \approx \beta v_t - \alpha \nabla f\left(\frac{x_t + \beta v_t + x_t}{2}\right), \\ x_{t+1} = x_t + v_{t-1}. \end{cases} \tag{7}$$

Back to the implicit form, numerically it will be unconditional stable.